

ESP32 SECURITY SUITE

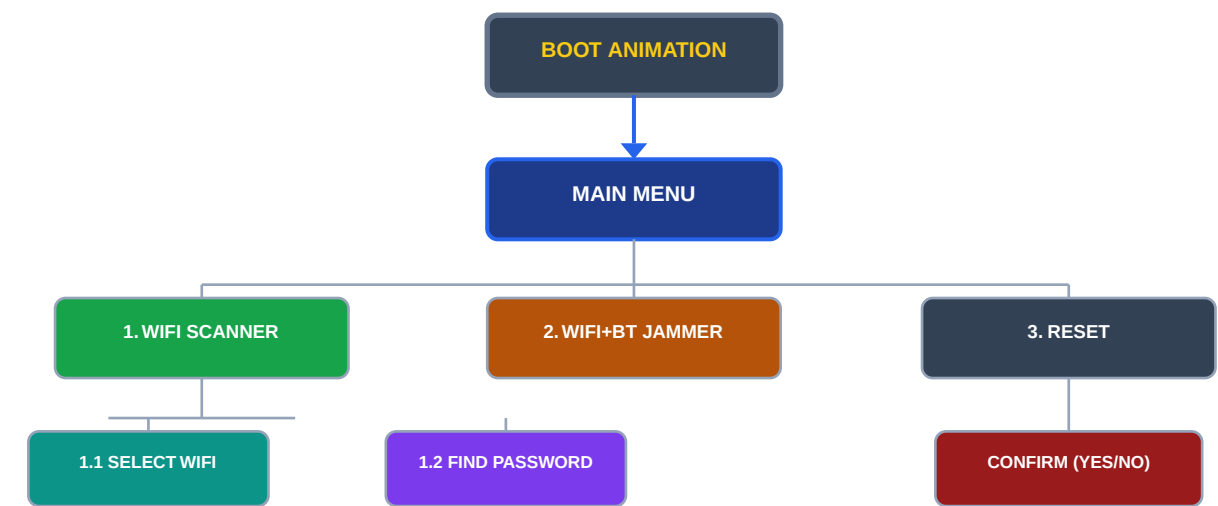
TFT Display UI Design Model — CradleGuard 3.0

2.4" MCUFRIEND TFT LCD | 320 × 240 px | Landscape

LEGAL DISCLAIMER:

This system is strictly for authorized security research and educational use in controlled environments.

Screen Navigation Flow



Navigation Controls:

BOOT button (GPIO 0) = GO BACK one level | Tap header "< ..." = GO BACK | Physical ON btn = context action

Screen Designs — Rendered Mockups

Screen 1: Main Menu

- Header: "CRADLEGUARD 3.0" (Blue bar)
- Button 1: WIFI SCANNER !' Dark Green !' GPIO 12 tap or touch
- Button 2: WIFI+BT JAMMER !' Dark Red !' Activates dual NRF24
- Button 3: RESET !' Slate Blue !' Calls ESP.restart()

Screen 2: Wi-Fi Scanner Sub-Menu

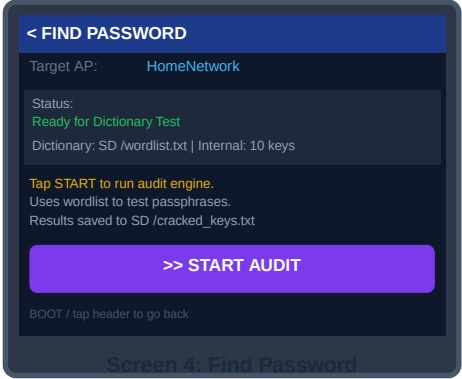
- "< WIFI SCANNER" header !' tap to go back
- Button 1.1: SELECT WIFI !' Teal !' Scans & lists APs
- Button 1.2: FIND PASSWORD !' Violet !' Runs wordlist audit
- BOOT button (GPIO 0) returns to Main Menu

Screen 3: 1.1 Select Wi-Fi Network

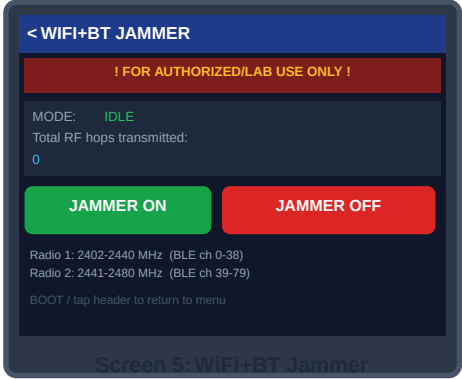
- Header "< SELECT WIFI" !' tap to return to sub-menu
- List rows: tap any row to highlight/select that network
- Highlighted row !' filled Cyan, text dark
- SCAN button !' Green, triggers scanWiFiNetworks()
- CONFIRM CHOICE !' Cyan, saves selectedTargetIndex & goes back
- Physical TURN ON (GPIO 12) also triggers a new scan

Screen Designs — Diagram Mockups

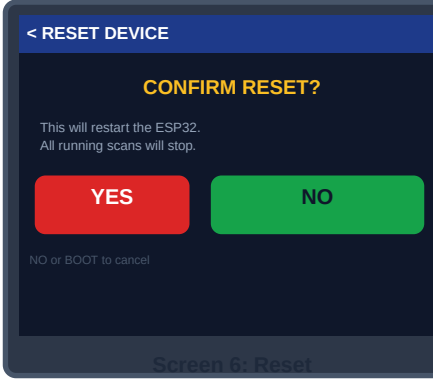
Screen 4: 1.2 Find Password / Dictionary Audit



- Tap START AUDIT !' calls runDictionaryCrack()
- Result shows in Status box (green = found)
- cracked_keys.txt written to SD card



- JAMMER ON !' actionActive = true, LED on
- JAMMER OFF !' actionActive = false
- RF hops counter updates live every 500ms



- YES !' ESP.restart() after 1.5s delay
- NO / BOOT !' back to Main Menu
- Displays "RESTARTING..." before reset

Touch Zone Map & Control Reference

Screen	Touch Zone / Action	ESP32 Event Triggered	GPIO / Button
Main Menu	Tap WIFI SCANNER row	setScreen(SCREEN_WIFI_SUBMENU)	Touch
Main Menu	Tap WIFI+BT JAMMER row	setMode(BLE) + setScreen(SCREEN_JAMMER)	Touch
Main Menu	Tap RESET row	setScreen(SCREEN_RESET)	Touch
WiFi Sub	Tap 1.1 SELECT WIFI	setScreen(SCREEN_WIFI_SELECT) + scan	Touch
WiFi Sub	Tap 1.2 FIND PASSWORD	setMode(WIFI_CRACK) + crack screen	Touch
WiFi Sub	Tap header "<"	setScreen(SCREEN_MAIN_MENU)	Touch / BOOT btn
Select WiFi	Tap a network row	selectedTargetIndex = i; redraw	Touch
Select WiFi	Tap >> SCAN	scanWiFiNetworks(); redraw list	Touch / GPIO 12
Select WiFi	Tap CONFIRM CHOICE	setScreen(SCREEN_WIFI_SUBMENU)	Touch
Find Password	Tap >> START AUDIT	runDictionaryCrack(); redraw	Touch / GPIO 12
Find Password	Tap header "<"	setScreen(SCREEN_WIFI_SUBMENU)	Touch / BOOT btn
Jammer	Tap JAMMER ON	actionActive = true; LED HIGH	Touch / GPIO 12
Jammer	Tap JAMMER OFF	actionActive = false; LED LOW	Touch / GPIO 14
Jammer	Tap header "<"	setScreen(SCREEN_MAIN_MENU)	Touch / BOOT btn
Reset	Tap YES	delay(1500) + ESP.restart()	Touch
Reset	Tap NO	setScreen(SCREEN_MAIN_MENU)	Touch / BOOT btn

Physical Button Mapping

Button	GPIO Pin	Wiring	Context Action
TURN ON Button	GPIO 12	One leg !' GPIO 12, other !' GND (INPUT_PULLUP)	Scan / Start Audit / Jammer ON
TURN OFF Button	GPIO 14	One leg !' GPIO 14, other !' GND (INPUT_PULLUP)	Jammer OFF / Stop Action
BOOT Button	GPIO 0	Built-in ESP32 BOOT pushbutton	Navigate BACK one screen level
Status LED	GPIO 2	Built-in ESP32 LED	HIGH when jammer/action is active

UI Color Palette Reference

Deep Navy (BG) Screen background	Header Blue Top header bars	Dark Green WIFI SCANNER button / JAMMER ON
Dark Red WIFI+BT JAMMER button / JAMMER OFF	Teal SELECT WIFI button	Violet FIND PASSWORD / START AUDIT button
Slate RESET button background	Cyan Accent Selected items / CONFIRM button	Yellow Warning Warning bars and instruction text

MicroSD Card UI & Dynamic Theme Engine

DYNAMIC SD-CARD DISPLAY SYSTEM (FAT32)

The firmware integrates an uncompressed 24-bit/16-bit BMP decoder, dynamic RGB565 theme parser, and timed lyrics/banner animator that reads directly from the MicroSD card slot on the 2.4" TFT shield (CS: GPIO 13, VSPI bus).

Supported SD Card Assets & Configuration Files:

File Name	Format / Spec	UI Function / Trigger	Fallback Behavior
/splash.bmp	320×240 24-bit BMP	Boot Splash Screen with progress bar	3D Wireframe Cube Only
/theme.txt	Key=Value (RGB565)	Dynamic palette, titles & badge toggles	Cyberpunk Dark Palette
/lyrics.txt	[mm:ss.xx] String	Live scrolling status / lyrics ticker	Built-in status messages
/menu_bg.bmp	320×240 24-bit BMP	Custom background wallpaper for menus	Solid Navy Background
/wordlist.txt	Plaintext passphrases	WPA2 offline dictionary cracker	Internal 10-key wordlist
/cracked_keys.txt	Append Log Text	Auto-exports found passwords & SSIDs	Serial CLI Output Only

Sample /theme.txt Configuration Format:

```
# CRADLEGUARD 3.0 SD THEME CONFIGURATION
THEME_NAME=Cyberpunk Neon
COLOR_BG=0x0821      # Deep Navy Background
COLOR_HEADER=0x001F  # Header Bar Blue
COLOR_CARD=0x18E3    # Container Card Gray
COLOR_TEXT=0xFFFF    # Primary Text (White)
COLOR_ACCENT=0x07FF  # Highlight / Accent (Cyan)
COLOR_GREEN=0x07E0   # Wifi Scanner / Confirm (Green)
COLOR_RED=0xF800     # Jammer / Deauth (Red)
APP_TITLE=CRADLEGUARD 3.0
SHOW_SD_BADGE=1      # [SD:OK] Header Indicator
USE_SD_SPLASH=1      # Render /splash.bmp on boot
```